

# fit\_sjsdm\_regularization\_candidate.R

August 10, 2026

## Contents

fit\_sjsdm\_regularization\_candidate . . . . . 1

**Index** 3

---

fit\_sjsdm\_regularization\_candidate  
*Fit One sjSDM Regularization Candidate*

---

## Description

Fits one prepared training fold with regularization parameters from a deterministic candidate table.

## Usage

```
fit_sjsdm_regularization_candidate(  
  data_train_input = NULL,  
  candidate = NULL,  
  sel_abiotic_formula = NULL,  
  config_model_fitting = NULL,  
  seed = 900723L,  
  device = "cpu",  
  biotic = NULL,  
  fit_function = fit_jsdm_model  
)
```

## Arguments

|                      |                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------|
| data_train_input     | Fold-local training input returned by [prepare_sjsdm_cross_validation_fold()].                 |
| candidate            | One-row candidate table containing ‘candidate_id’ and the six sjSDM regularization parameters. |
| sel_abiotic_formula  | Abiotic model formula passed to [fit_jsdm_model()].                                            |
| config_model_fitting | Active model-fitting configuration list.                                                       |
| seed                 | Deterministic integer seed for the candidate fit.                                              |

|                           |                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>device</code>       | Device passed to <code>[fit_jsdm_model()]</code> . Defaults to <code>"cpu"</code> .                                                                                                                                                          |
| <code>biotic</code>       | Optional sjSDM biotic structure. <code>'NULL'</code> retains the regularized covariance structure created by <code>[fit_jsdm_model()]</code> . A diagonal structure disables residual cross-taxon associations for predictive decomposition. |
| <code>fit_function</code> | Injectable fit function. Defaults to <code>[fit_jsdm_model()]</code> .                                                                                                                                                                       |

**Details**

The adapter keeps regularization forwarding in one tested place so tuning and selected-candidate reruns use identical fitting arguments.

**Value**

Fitted model object returned by `'fit_function'`.

# Index

`fit_sjsdm_regularization_candidate`, [1](#)